APPENDIX II—UPDATE OF THE DEBT SUSTAINABILITY ANALYSIS (DSA) USING THE LOW-INCOME COUNTRY FRAMEWORK[1]

The analysis based on the joint IMF-World Bank debt sustainability framework for low-income countries shows that Togo is at moderate risk of debt distress*. After full HIPC assistance, MDRI and beyond HIPC assistance, Togo's external and public debt indicators have improved significantly and are projected remain below the relevant thresholds in the 20-year period under baseline assumptions, thanks to a stable economic and political climate and rehabilitation of key sectors. However, Togo remains vulnerable to certain shocks and could breach the policy-related thresholds for the PV of debt-to-GDP and the PV of debt-to-revenue ratios under some alternative scenarios in the latter years.*

A. Introduction

1. **The last joint DSA for Togo was prepared in November 2010 and concluded that Togo was at moderate risk of debt distress after full HIPC assistance, MDRI and beyond HIPC assistance.** This debt sustainability analysis (LIC-DSA) for Togo assesses its external and public debt using the forward-looking debt sustainability framework (DSF) for low-income countries.[2] The outcome of this analysis was in line with previous DSAs, finding generally favorable debt dynamics as a consequence of expected enhancements in the macroeconomic policy framework, notably fiscal discipline and solid implementation of growth-promoting structural reforms. Togo reached the completion point of the HIPC Initiative in December 2010 and was granted debt relief from several multilateral and bilateral creditors. As a result, the nominal debt stock fell from $1.7 billion at end-2009 to about $0.5 billion at end-2010.

2. **The authorities are continuing to make good faith efforts toward reaching agreements with non-Paris Club bilateral creditors, as well as other official and commercial creditors on the provision of debt relief on terms comparable to the relief accorded in the context of the HIPC initiative**. At the time of HIPC completion, 88 percent of the multilateral HIPC assistance had already been fully delivered through arrears clearance and

---

[1] This Appendix was prepared by Antonio C. David (AFR).

[2] This DSA update has been prepared by Fund staff using the Debt Sustainability Framework (DSF) for Low Income Countries (see "Applying the Debt Sustainability Framework for Low-Income Countries Post Debt Relief", (http://siteresources.worldbank.org/INTDEBTDEPT/PolicyPapers/21154573/DMSDR1S3149398v1DSFPaperforweb.pdf). Togo's quality of policies and institutions, as measured by the average World Bank's Country Policy and Institutional Assessment (CPIA) for the period 2008-2010 (2.8), places it as a "weak performer". The corresponding indicative thresholds for the external debt indicators are 30 percent for the NPV of debt-to-GDP ratio, 100 percent of the debt-to-export ratio, 200 percent for NPV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio, and 25 percent for the debt service-to-revenue ratio.

concessional rescheduling of arrears. The World Bank and AfDB have provided their share of debt relief under MDRI after the completion point. Paris Club creditors have also provided relief in December 2010, effectively leading to a 95 percent reduction of Togolese debt towards these creditors. China delivered its share of relief through a 2007 debt cancellation, as did Kuwait through a 2009 loan rescheduling agreement.

3. **In the LIC-DSA framework, the present value of Togo's public and publicly guaranteed (PPG) external debt is $539 million at end-2010, after full delivery of HIPC, MDRI and beyond HIPC assistance.** Around 56 percent of this stock of debt is owed to multilateral creditors and the remainder to bilateral and commercial creditors.

B. Baseline Assumptions

4. **The baseline macroeconomic assumptions for the present DSA are the following:**

a. Real GDP growth is projected to exceed its estimated long-run growth trend of around 4 percent in the period from 2012 to 2016, largely due to increases in public investment and the fiscal deficit (see below). Subsequently, it will vary around 4 percent, driven by the ongoing rehabilitation of the phosphate and cotton sectors, growth in the agricultural sector (especially food production), an improved investment climate, increased financial intermediation; additional FDI and foreign aid; and growing regional integration, thanks to Togo's strategic geographical location and the role of the port of Lomé. The 4 percent potential growth was estimated based on a population growth rate of 2.5 percent (which reflects the latest available, finalized data) and by building-up sectoral growth rates under the assumptions outlined above and planned growth-enhancing reforms. The estimated potential growth rate exceeds historical rates, which were depressed by the dislocations caused by the protracted social, political, and economic crisis that the country experienced up to the mid-2000s.

b. Public investment is projected to accelerate starting in 2012, reaching 12.6 percent of GDP in 2013 and 2014 and then reverting back to levels close to the ones observed in 2011. About 35 percent of this increase will be financed by loans (75 percent of which will be concessional loans). Public investment projects are expected to be mostly directed to infrastructure and will lead GDP growth to increase above its long-term trend for these years.

c. The projections for key commodity prices (oil, cotton, cocoa, and coffee) through 2016 are based on WEO projections of May 2011 and are assumed constant in real terms afterwards.

d. Inflation over the long-term is projected to remain stable at around 2 percent, reflecting sound monetary policy at the regional level.

e. There is a gradual improvement in the current account deficit over the medium-term, following a deterioration in the early projection years. The deficit remains high, around 5 percent of GDP on average between 2017 and 2031, with higher exports of phosphates, cement and clinker being insufficient to fully compensate for the growth in imports as foreign aid is absorbed and foreign investment increases.

f. The domestic primary fiscal deficit is assumed to stay close to zero[3] during the projected period, thereby providing a fiscal anchor to ensure fiscal sustainability over the long term. The scaling up of public investment in 2012-14 is projected to lead to a temporary deviation from this fiscal balance

g. FDI and donor flows are expected to increase over the medium-term reflecting improvements in the investment climate and overall governance.

h. External financing is initially mostly on grant[4] and highly concessional terms, with less concessional financing gradually picking up, leading to a decrease in the grant element of new financing (as shown in Figure 1). Concessional external financing is expected to come largely from Togo's traditional multilateral donors.

C. External Debt Sustainability Analysis

Baseline

5. **Under the baseline scenario, Togo's external debt indicators remain below their relevant indicative thresholds (Table 1a, Figure 1).** The present value (PV) of public and publicly guaranteed (PPG) debt equals to 15.2 percent of GDP in 2011 and remains below the 30 percent threshold until the end of the projected period. Furthermore, both the PV of external debt relative to revenues and exports stay below their respective indicative threshold until 2031. Nevertheless, the three PV ratios (under the baseline scenario) increase significantly over the projected period. This result reflects in part conservative assumptions concerning the outer years, particularly a declining share of concessional loans in new financing, which results in the concessionality of new loans decreasing from 35 percent to close to 5 percent over the period

[3] The primary deficit is projected to stay at around 2 percent of GDP over the projection period, with the temporary exception of 2012-14.

[4] The proportion of financing needs covered by grants declines from 80 percent at the beginning of the projection period to less than 50 percent towards the end.

2011 to 2031. In fact, as shown in Table 1a, the contribution of the nominal interest rate to endogenous debt dynamics increases significantly in the post-2016 period[5].

Alternative Scenarios and Stress Tests

6. **Togo's external debt outlook remains vulnerable to shocks especially toward the end of the projection period** (Table 1b, Figure 1). The policy thresholds for two key ratios are breached under the most extreme stress tests. The PV of external debt to GDP indicator deteriorates significantly under the scenario in which new loans are obtained on less favorable terms. In this case, the threshold is breached in 2024. In addition, the PV of debt-to-revenue ratios also surpasses the indicative thresholds at the end of the period under the "less favorable terms" shock.

7. **In light of the results from the baseline and alternative scenarios as well as the stress tests, IMF staff concludes that Togo still is at moderate risk of debt distress.**

D. Public Sector Debt Sustainability

Baseline

8. **The inclusion of Togo's domestic public debt in the analysis emphasizes the vulnerabilities of the baseline scenario (Table 2a, Figure 2).** Togo's domestic debt burden reflects years of weak fiscal management and domestic arrears accumulation, as well as the need to recapitalize ailing banks. Much of this debt arose from losses at state-owned enterprises in the past, highlighting the need for further reforms in the governance of these enterprises. Under the baseline scenario, the ratios of total public debt in present value terms are projected to decrease until 2018 before picking up gradually towards the end of the period. These trends are linked to the assumptions regarding the concessionality of financing, with the share of grants gradually falling and increased financing from the regional financial markets. On average, the PV of debt to GDP ratio amounts to around 23.2 percent over the overall period. Given the assumed improvement in the macroeconomic outlook and the cautious debt strategy in the baseline scenario, debt ratios would remain at a reasonable level over the long-run.

Alternative Scenarios and Stress Tests

9. **The evolution of the debt indicators would be highly sensitive to the most extreme shock--a lower growth rate—which would increase the debt level and debt service over the long run**. Total public debt dynamics are particularly vulnerable to a growth shock, which would

[5] The substantial residual for endogenous debt dynamics presented in Table 1a for 2011 is largely due to exceptional financing, price and exchange rate changes, and debt flow relief. Debt flow relief and price and exchange rate changes also account for over half of the residuals observed in subsequent years.

lead the PV of debt to GDP ratio to break the 30 percent threshold by 2014 (Table 2b, Figure 2). This highlights the importance of a reform agenda that improves the business environment to support foreign investment and growth.[6]

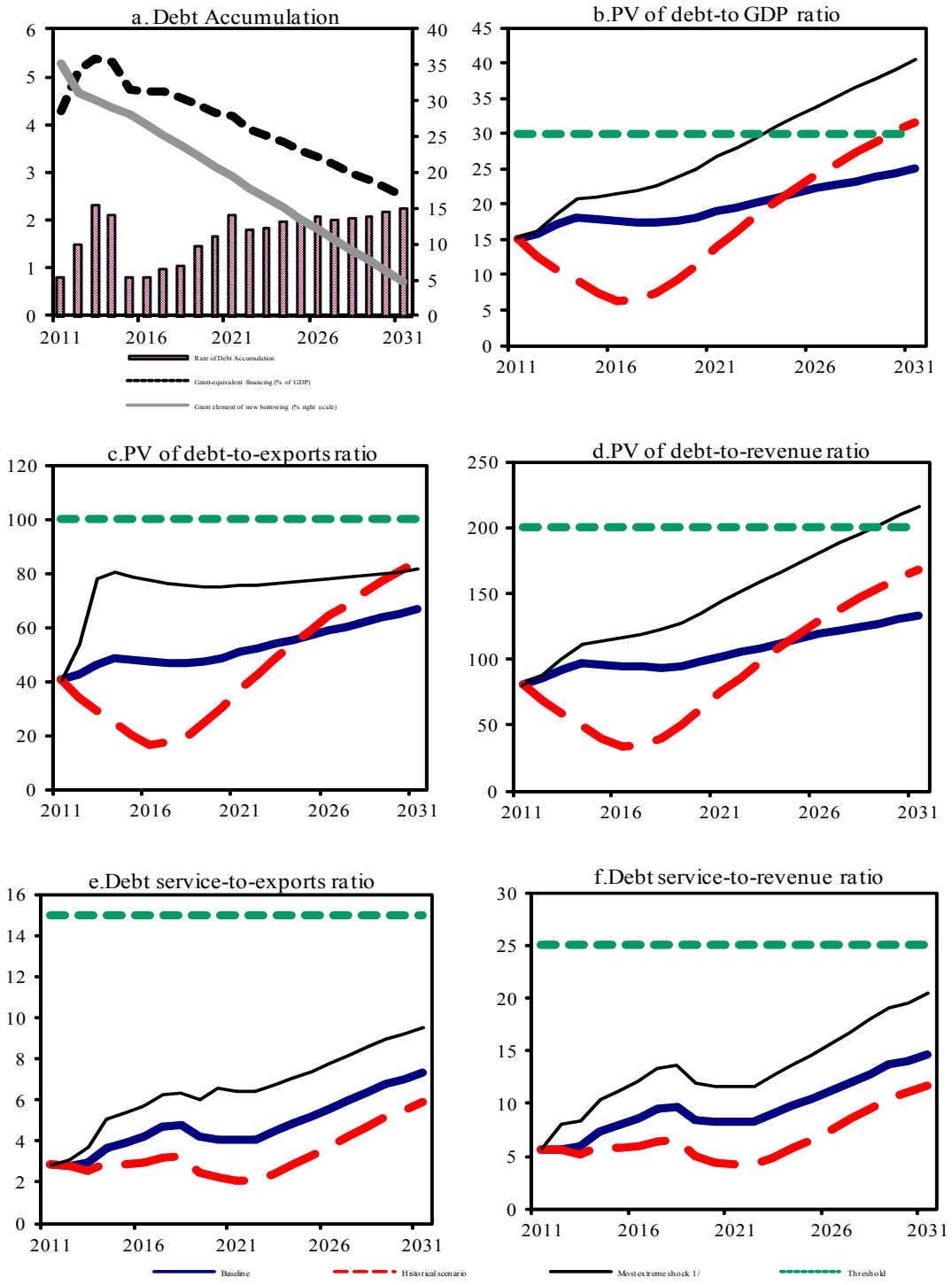
E. Conclusion

10. **The DSA indicates that Togo is at moderate risk of debt distress after reaching the completion point.** Under the baseline scenario, the debt ratios remain below the thresholds for the projected period, but under less positive yet realistic alternative scenarios, certain thresholds would be breached in the latter years. These dynamics highlight the vulnerability of the Togolese economy to certain shocks (in particular less favorable terms on new financing) and stress the need for a prudent approach to new borrowing and continuing to seek grant and highly concessional financing to the extent possible, as well as pursuing reforms to improve debt management capacity. The assumptions and conclusions of the DSA were discussed with the authorities, who broadly concurred.

11. **Maintaining a robust external debt outlook will depend on a sustained pick-up in real GDP growth and foreign direct investment, as well as prudent debt management and solid fiscal performance.** The inclusion of Togo's domestic debt in the analysis reinforces the conclusions of the external DSA and stresses the risks to Togo's debt prospects. In this context, it is essential that the Togolese authorities continue current efforts, particularly to strengthen public financial management (especially debt management), restructure the banking system, reform state-owned enterprises, and improve the investment climate hence laying the foundation for accelerating growth.

[6] Note the historical scenario assumes that real GDP growth and the primary balance are set at historical averages (over the last 10 years). In the case of Togo, this yields a primary surplus 0.6 percent of GDP for the historical scenario, compared to a primary deficit in the baseline. As a result, gross financing needs and therefore gross new borrowing under the historical scenario are much lower than under the baseline. The large negative new borrowing compared to the baseline, eventually leads to negative PV ratios.

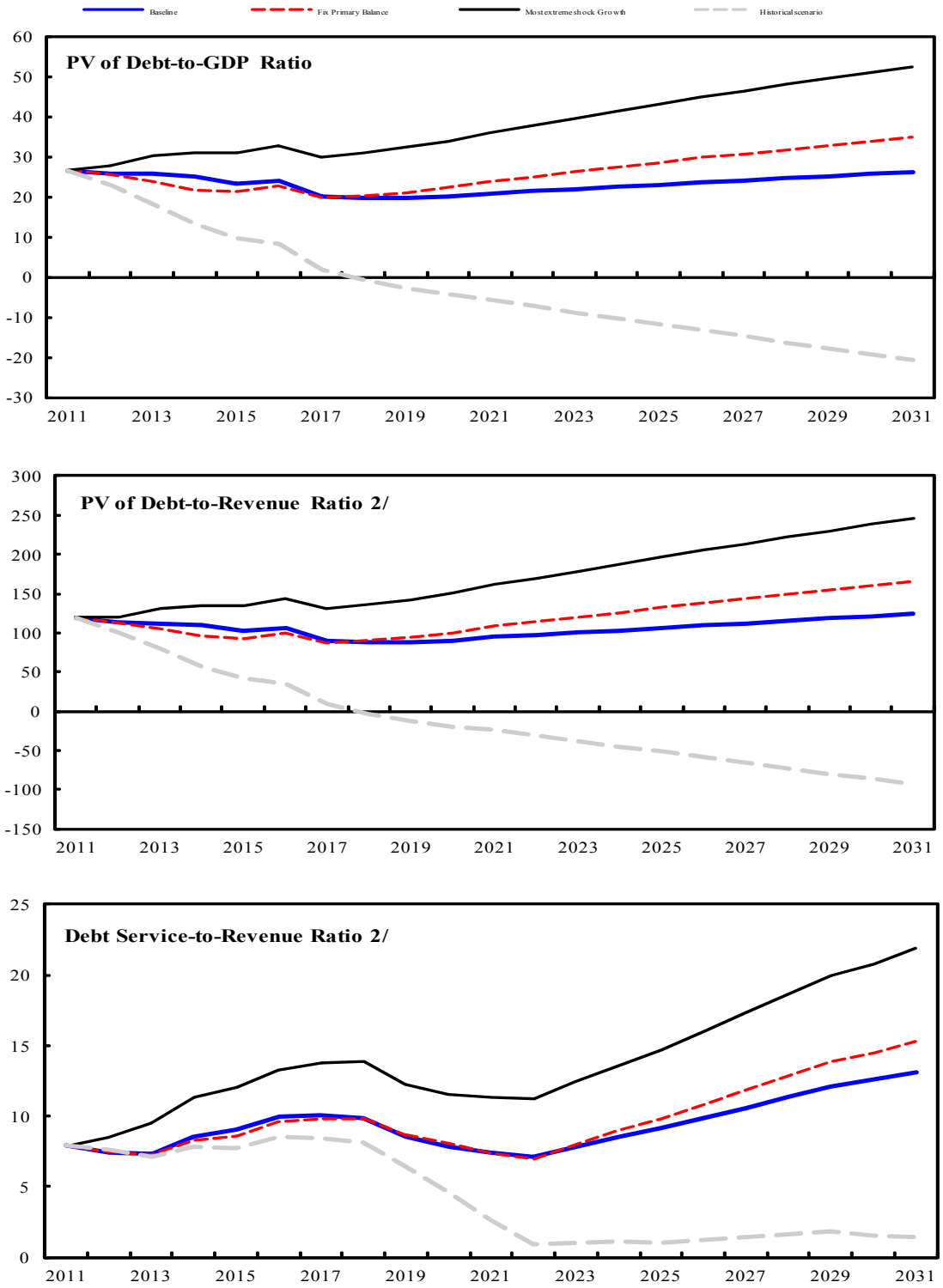
Figure 1. Togo: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2021. In figure b. it corresponds to a Terms shock; in c. to a Exports shock; in d. to a Terms shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2.Togo: Indicators of Public Debt Under Alternative Scenarios, 2011-2031 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2021.
2/ Revenues are defined inclusive of grants.

Table 1a.: External Debt Sustainability Framework, Baseline Scenario, 2008-2031 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2008	2009	2010			2011	2012	2013	2014	2015	2016	2011-2016 Average	2021	2031	2017-2031 Average
External debt (nominal) 1/	**61.6**	**53.9**	**19.1**			**18.2**	**19.6**	**22.1**	**24.0**	**24.2**	**24.3**		**27.4**	**33.2**	
o/w public and publicly guaranteed (PPG)	60.9	52.7	17.2			16.0	17.1	19.2	20.9	20.8	20.7		23.3	29.5	
Change in external debt	-17.5	-7.6	-34.8			-0.9	1.5	2.5	1.9	0.2	0.1		1.2	0.4	
Identified net debt-creating flows	-10.6	6.5	6.2			6.0	6.3	6.3	5.7	4.8	4.2		1.7	1.3	
Non-interest current account deficit	**6.4**	**6.3**	**6.8**	**6.4**	**0.5**	**7.3**	**7.8**	**8.0**	**7.8**	**6.9**	**6.3**		**4.4**	**3.7**	4.2
Deficit in balance of goods and services	16.4	15.6	16.5			17.7	18.6	18.9	18.4	17.2	16.5		14.5	13.9	
Exports	35.5	36.8	37.3			37.4	37.1	37.2	37.2	37.3	37.4		37.5	37.4	
Imports	51.9	52.5	53.8			55.1	55.8	56.0	55.6	54.5	53.9		52.0	51.3	
Net current transfers (negative = inflow)	-10.1	-9.6	-10.2	-9.6	1.7	-11.0	-11.5	-11.5	-11.3	-11.0	-11.0		-10.6	-10.2	-10.5
o/w official	-1.4	-1.5	-2.0			-3.4	-4.3	-4.2	-4.2	-4.1	-4.1		-3.5	-2.3	
Other current account flows (negative = net inflow)	0.0	0.3	0.4			0.6	0.6	0.6	0.7	0.7	0.8		0.4	0.0	
Net FDI (negative = inflow)	**-1.3**	**-0.4**	**-0.6**	**-2.8**	**1.7**	**-1.0**	**-1.0**	**-1.2**	**-1.5**	**-1.5**	**-1.5**		**-2.2**	**-2.0**	-2.1
Endogenous debt dynamics 2/	**-15.8**	**0.6**	**0.1**			**-0.3**	**-0.4**	**-0.5**	**-0.6**	**-0.6**	**-0.6**		**-0.5**	**-0.4**	
Contribution from nominal interest rate	0.4	0.4	0.4			0.3	0.3	0.3	0.4	0.4	0.4		0.6	0.9	
Contribution from real GDP growth	-1.5	-2.0	-2.0			-0.6	-0.8	-0.9	-1.0	-1.0	-1.0		-1.0	-1.3	
Contribution from price and exchange rate changes	-14.7	2.2	1.7			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**-6.9**	**-14.2**	**-41.1**			**-7.0**	**-4.9**	**-3.8**	**-3.8**	**-4.6**	**-4.1**		**-0.5**	**-0.9**	
o/w exceptional financing	-2.3	-2.8	-2.6			-2.6	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	18.9			17.4	18.3	19.9	21.1	21.2	21.2		23.1	28.8	
In percent of exports	...	...	50.6			46.4	49.2	53.6	56.8	56.7	56.6		61.6	76.8	
PV of PPG external debt	**...**	**...**	**17.0**			**15.2**	**15.8**	**17.1**	**18.0**	**17.8**	**17.6**		**19.0**	**25.0**	
In percent of exports	**...**	**...**	**45.5**			**40.6**	**42.4**	**45.9**	**48.3**	**47.6**	**47.0**		**50.7**	**66.8**	
In percent of government revenues	**...**	**...**	**90.1**			**80.5**	**85.6**	**92.2**	**97.3**	**96.1**	**94.9**		**102.6**	**133.6**	
Debt service-to-exports ratio (in percent)	**1.2**	**4.6**	**5.1**			**2.9**	**2.8**	**2.9**	**3.7**	**3.9**	**4.3**		**4.1**	**7.3**	
PPG debt service-to-exports ratio (in percent)	**1.2**	**4.6**	**5.1**			**2.9**	**2.8**	**2.9**	**3.7**	**3.9**	**4.3**		**4.1**	**7.3**	
PPG debt service-to-revenue ratio (in percent)	**2.8**	**10.0**	**10.1**			**5.7**	**5.7**	**5.9**	**7.4**	**7.9**	**8.6**		**8.2**	**14.7**	
Total gross financing need (Billions of U.S. dollars)	0.2	0.3	0.3			0.3	0.3	0.4	0.4	0.4	0.4		0.4	0.8	
Non-interest current account deficit that stabilizes debt ratio	23.9	13.9	41.6			8.3	6.3	5.6	5.9	6.7	6.2		3.2	3.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.4	3.2	3.7	2.1	2.1	3.9	4.5	4.7	4.9	4.5	4.4	4.5	4.1	4.2	4.1
GDP deflator in US dollar terms (change in percent)	22.8	-3.5	-3.1	7.4	8.2	11.5	2.8	1.1	1.5	1.2	1.5	3.3	2.3	2.2	2.3
Effective interest rate (percent) 5/	0.7	0.6	0.7	1.6	0.7	2.0	1.9	1.9	1.8	1.8	1.8	1.9	2.2	3.1	2.4
Growth of exports of G&S (US dollar terms, in percent)	22.2	3.4	1.9	10.9	12.9	16.2	6.6	6.1	6.6	6.1	6.0	7.9	6.7	6.4	6.5
Growth of imports of G&S (US dollar terms, in percent)	26.9	0.7	3.1	12.8	17.9	18.7	8.7	6.5	5.7	3.6	4.8	8.0	6.7	6.3	6.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	35.2	31.1	30.1	29.1	28.1	26.7	30.0	19.4	4.8	15.0
Government revenues (excluding grants, in percent of GDP)	15.6	16.9	18.9			18.9	18.4	18.5	18.5	18.5	18.5	18.5	18.5	18.7	18.6
Aid flows (in Billions of US dollars) 7/	0.0	0.1	0.1			0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4
o/w Grants	0.0	0.0	0.1			0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3
o/w Concessional loans	0.0	0.0	0.1			0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.3	5.1	5.4	5.3	4.7	4.7	4.9	4.2	2.5	3.6
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			72.2	73.2	66.8	66.0	75.4	73.4	71.2	59.5	39.0	55.0
Memorandum items:															
Nominal GDP (Billions of US dollars)	3.2	3.2	3.2			3.7	4.0	4.2	4.5	4.7	5.0	4.3	6.9	12.8	8.6
Nominal dollar GDP growth	25.7	-0.3	0.5			15.9	7.4	5.9	6.5	5.7	6.0	7.9	6.5	6.5	6.5
PV of PPG external debt (in Billions of US dollars)			0.5			0.6	0.6	0.7	0.8	0.8	0.9	0.7	1.3	3.2	1.8
(PVt-PVt-1)/GDPt-1 (in percent)						0.8	1.5	2.3	2.1	0.8	0.8	1.4	2.1	2.2	1.8
Gross workers' remittances (Billions of US dollars)	0.2	0.3	0.3			0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.5	1.0	0.6
PV of PPG external debt (in percent of GDP + remittances)	...	...	15.7			14.1	14.7	15.9	16.8	16.6	16.4	15.8	17.7	23.2	19.4
PV of PPG external debt (in percent of exports + remittances)	...	...	37.4			33.7	35.6	38.4	40.6	40.2	39.6	38.0	42.6	55.1	46.4
Debt service of PPG external debt (in percent of exports + remittances)	...	...	4.2			2.4	2.4	2.5	3.1	3.3	3.6	2.9	3.4	6.0	4.4

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b.Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031
(In percent)

	Projections							
	2011	2012	2013	2014	2015	2016	**2021**	2031
PV of debt-to GDP ratio								
Baseline	15	16	17	18	18	18	**19**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	15	13	11	9	7	6	**14**	32
A2. New public sector loans on less favorable terms in 2011-2031 2	15	16	19	21	21	21	**27**	40
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	15	16	19	20	19	19	**20**	27
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	15	18	25	25	25	25	**24**	26
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	15	16	18	19	19	18	**20**	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	15	19	23	24	24	23	**23**	26
B5. Combination of B1-B4 using one-half standard deviation shocks	15	19	24	25	24	24	**24**	27
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	15	22	24	25	25	25	**26**	34
PV of debt-to-exports ratio								
Baseline	41	42	46	48	48	47	**51**	67
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	41	34	29	25	20	17	**37**	84
A2. New public sector loans on less favorable terms in 2011-2031 2	41	43	50	56	56	57	**71**	108
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	41	42	46	48	47	47	**50**	65
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	41	54	78	80	79	77	**76**	81
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	41	42	46	48	47	47	**50**	65
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	41	50	63	65	63	62	**62**	68
B5. Combination of B1-B4 using one-half standard deviation shocks	41	50	63	65	64	63	**63**	71
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	41	42	46	48	47	47	**50**	65
PV of debt-to-revenue ratio								
Baseline	81	86	92	97	96	95	**103**	134
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	81	68	58	50	40	34	**75**	168
A2. New public sector loans on less favorable terms in 2011-2031 2	81	88	100	112	114	116	**144**	216
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	81	89	100	106	104	103	**110**	143
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	81	100	133	137	135	133	**130**	139
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	81	88	97	102	101	99	**107**	138
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	81	102	126	130	128	126	**125**	137
B5. Combination of B1-B4 using one-half standard deviation shocks	81	101	128	133	131	129	**128**	142
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	81	120	129	136	135	133	**142**	184

Table 1b.Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2011-2031 (continued)
(In percent)

Debt service-to-exports ratio

Baseline	3	3	3	4	4	4	**4**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	3	3	3	3	3	3	**2**	6
A2. New public sector loans on less favorable terms in 2011-2031 2	3	3	3	4	4	4	**4**	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	3	3	3	4	4	4	**4**	7
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	3	3	4	5	5	6	**6**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	3	3	3	4	4	4	**4**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	3	3	3	4	4	5	**5**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	3	4	5	5	**5**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	3	3	3	4	4	4	**4**	7

Debt service-to-revenue ratio

Baseline	6	6	6	7	8	9	**8**	15
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2011-2031 1/	6	6	5	6	6	6	**4**	12
A2. New public sector loans on less favorable terms in 2011-2031 2	6	6	6	7	8	8	**8**	19
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2012-2013	6	6	6	8	9	9	**9**	16
B2. Export value growth at historical average minus one standard deviation in 2012-2013 3/	6	6	6	9	9	10	**11**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2012-2013	6	6	6	8	8	9	**9**	15
B4. Net non-debt creating flows at historical average minus one standard deviation in 2012-2013 4/	6	6	6	8	9	10	**11**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	6	6	7	9	9	10	**11**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2012 5/	6	8	8	10	11	12	**12**	20
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	9	9	9	9	9	9	**9**	9

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Togo: Public Sector Debt Sustainability Framework, Baseline Scenario, 2008-2031
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2008	2009	2010	Average	Standard Deviation	2011	2012	2013	2014	2015	2016	2011-16 Average	2021	2031	2017-31 Average
Public sector debt 1/	83.1	67.8	32.3			27.4	27.1	27.8	27.8	26.3	27.0		25.1	30.6	
o/w foreign-currency denominated	60.9	52.7	17.2			16.0	17.1	19.2	20.9	20.8	20.7		23.3	29.5	
Change in public sector debt	-22.5	-15.3	-35.5			-4.9	-0.3	0.7	0.0	-1.5	0.7		1.1	0.5	
Identified debt-creating flows	-9.3	-5.7	3.3			0.3	2.5	3.9	3.9	1.2	1.4		1.2	1.3	
Primary deficit	0.1	1.9	0.6	-0.6	1.8	3.0	3.2	4.7	4.7	1.8	1.9	3.2	2.1	2.2	2.1
Revenue and grants	17.0	18.5	20.9			22.2	22.7	22.7	22.7	22.6	22.6		22.0	21.0	
of which: grants	1.4	1.5	2.0			3.4	4.3	4.2	4.2	4.1	4.1		3.5	2.3	
Primary (noninterest) expenditure	17.1	20.3	21.5			25.2	25.9	27.4	27.4	24.5	24.5		24.1	23.2	
Automatic debt dynamics	-9.4	-7.6	2.7			-2.7	-0.7	-0.7	-0.9	-0.6	-0.5		-0.9	-0.9	
Contribution from interest rate/growth differential	-14.7	-3.1	-2.6			-1.4	-0.9	-0.9	-1.0	-0.7	-0.6		-0.9	-0.9	
of which: contribution from average real interest rate	-12.2	-0.5	-0.1			-0.2	0.3	0.3	0.3	0.4	0.5		0.1	0.3	
of which: contribution from real GDP growth	-2.5	-2.6	-2.4			-1.2	-1.2	-1.2	-1.3	-1.2	-1.1		-1.0	-1.2	
Contribution from real exchange rate depreciation	5.3	-4.4	5.3			-1.3	0.2	0.2	0.2	0.1	0.1		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-13.2	-9.7	-38.8			-5.2	-2.8	-3.2	-3.8	-2.7	-0.7		-0.1	-0.8	
Other Sustainability Indicators															
PV of public sector debt	...	...	32.1			26.5	25.7	25.6	24.9	23.3	23.8		20.8	26.2	
o/w foreign-currency denominated	...	...	17.0			15.2	15.8	17.1	18.0	17.8	17.6		19.0	25.0	
o/w external	...	...	17.0			15.2	15.8	17.1	18.0	17.8	17.6		19.0	25.0	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	-3.1	4.1	2.7			4.7	4.9	6.3	6.7	3.9	4.1		3.7	5.0	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	153.3			119.4	113.2	112.5	109.9	102.9	105.5		94.5	124.3	
PV of public sector debt-to-revenue ratio (in percent)	…	…	169.9			140.8	139.6	138.3	134.7	125.8	128.7		112.3	139.7	
o/w external 3/	…	…	90.1			80.5	85.6	92.2	97.3	96.1	94.9		102.6	133.6	
Debt service-to-revenue and grants ratio (in percent) 4/	-18.8	12.2	10.4			7.9	7.5	7.3	8.6	9.0	10.0		7.4	13.2	
Debt service-to-revenue ratio (in percent) 4/	-20.5	13.3	11.5			9.3	9.2	9.0	10.5	11.1	12.1		8.8	14.8	
Primary deficit that stabilizes the debt-to-GDP ratio	22.6	17.2	36.1			7.9	3.5	4.0	4.7	3.3	1.2		1.0	1.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.4	3.2	3.7	2.1	2.1	3.9	4.5	4.7	4.9	4.5	4.4	4.5	4.1	4.2	4.1
Average nominal interest rate on forex debt (in percent)	0.7	0.6	0.7	1.6	0.7	2.1	2.1	2.0	2.0	1.9	1.9	2.0	2.4	3.3	2.6
Average real interest rate on domestic debt (in percent)	-11.3	1.0	2.6	-1.0	6.6	0.7	3.3	3.7	4.5	6.9	10.3	4.9	3.2	0.0	2.8
Real exchange rate depreciation (in percent, + indicates depreciation)	7.8	-7.6	10.5	-3.3	10.6	-8.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	14.4	1.9	1.8	3.5	6.4	3.8	2.7	2.4	2.6	2.0	2.2	2.6	2.3	2.2	2.3
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.1	0.2	0.1	0.1	0.1	0.2	0.1	0.1	0.0	-0.1	0.0	0.1	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	35.2	31.1	30.1	29.1	28.1	26.7	30.0	19.4	4.8	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b.Togo: Sensitivity Analysis for Key Indicators of Public Debt 2011-2031

	Projections							
	2011	2012	2013	2014	2015	2016	2021	2031
PV of Debt-to-GDP Ratio								
Baseline	27	26	26	25	23	24	21	26
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	27	23	18	13	10	8	-6	-21
A2. Primary balance is unchanged from 2011	27	25	24	22	21	23	24	35
A3. Permanently lower GDP growth 1/	27	26	26	26	24	25	26	42
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	27	28	30	31	31	33	36	52
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	27	24	21	20	19	19	17	23
B3. Combination of B1-B2 using one half standard deviation shocks	27	24	20	21	21	22	25	41
B4. One-time 30 percent real depreciation in 2012	27	32	31	30	28	28	24	31
B5. 10 percent of GDP increase in other debt-creating flows in 2012	27	35	34	33	32	32	28	31
PV of Debt-to-Revenue Ratio 2/								
Baseline	119	113	112	110	103	106	94	124
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	119	100	80	57	41	36	-24	-93
A2. Primary balance is unchanged from 2011	119	112	105	95	93	100	108	165
A3. Permanently lower GDP growth 1/	119	114	114	113	108	112	115	200
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	119	120	131	135	134	143	161	247
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	119	105	91	89	82	85	76	111
B3. Combination of B1-B2 using one half standard deviation shocks	119	104	88	92	90	98	113	193
B4. One-time 30 percent real depreciation in 2012	119	141	137	132	124	126	111	147
B5. 10 percent of GDP increase in other debt-creating flows in 2012	119	153	151	148	140	142	127	149
Debt Service-to-Revenue Ratio 2/								
Baseline	8	7	7	9	9	10	7	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	8	7	8	8	8	3	1
A2. Primary balance is unchanged from 2011	8	7	7	8	9	10	7	15
A3. Permanently lower GDP growth 1/	8	7	7	9	9	10	8	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2012-2013	8	8	8	10	10	11	10	21
B2. Primary balance is at historical average minus one standard deviations in 2012-2013	8	7	7	8	8	9	6	12
B3. Combination of B1-B2 using one half standard deviation shocks	8	8	7	8	9	10	7	17
B4. One-time 30 percent real depreciation in 2012	8	8	9	11	12	13	11	22
B5. 10 percent of GDP increase in other debt-creating flows in 2012	8	7	9	10	10	11	10	16

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.